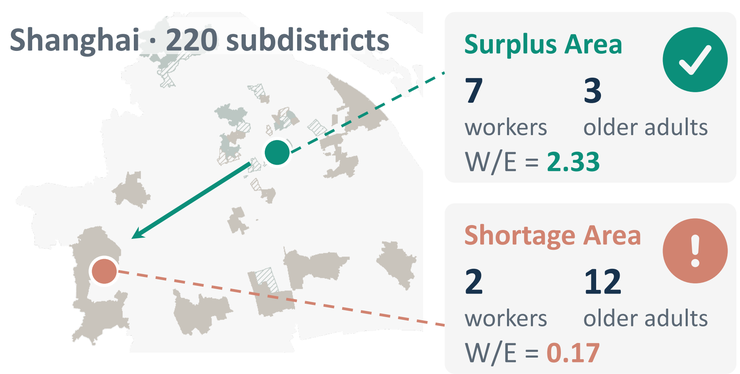
Uncovering the Spatial Trilemma of Cost, Equity, and Resilience in Community Worker Allocation

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01 BACKGROUND & MOTIVATION

Why need worker dispatch?

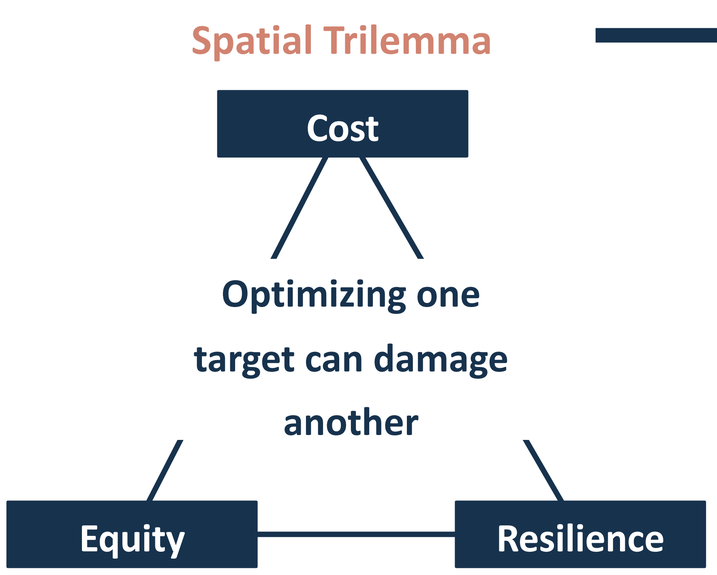


Examples: worker–elderly imbalance in two subdistricts

48,500 community workers | **84,301** older residents | **0.575** citywide W/E ratio

Worker–elderly mismatch requires spatial dispatch.

The trilemma directly motivates Tri-CORE



Tri-CORE Joint Formulation

One objective. Three priorities.

$$\min Z(\mathbf{x}) = \alpha \frac{C(\mathbf{x})}{C_{\text{ref}}} + \beta \frac{T(\mathbf{x})}{T_{\text{init}}} + \gamma \frac{R(\mathbf{x})}{R_{\text{init}}},$$

C(x):Cost **T(x):Equity** **R(x):Resilience**

α:β:γ makes policy priorities explicit

A configurable balance, not a single-objective optimum

Tri-CORE exposes the trade-off instead of hiding it.

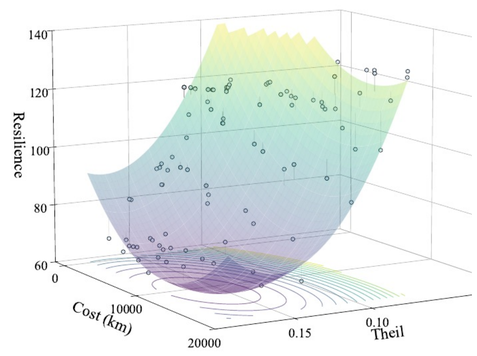
02 MAIN RESULTS

Tri-CORE balances all three targets

Method	Cost ↓ (km)	Equity ↓ Theil Index Δ%	Resilience ↓ R Δ%
Pure Cost	9,826	+97.7% ↑	−8.4% ↓
Pure Equity	23,089 ↑	−42.1% ↓	+19.6% ↑
Pure Resilience	7,335 ↓	+57.6% ↑	−41.9% ↓
Tri-CORE	**9,854**	**−21.4%**	**−16.9%**

−21.4% inequality · −16.9% backup deficit · only+0.3% distance

Parameter analysis: no free optimum



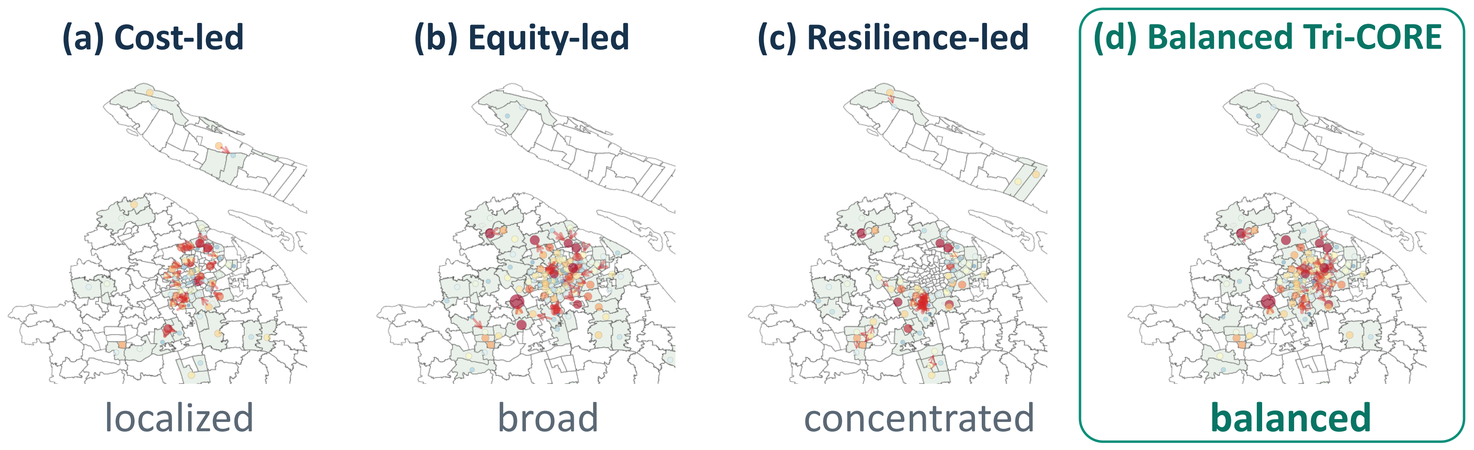
98 weight sets | **R^2 = 0.9664** quadratic fit

- Equity–resilience is the dominant conflict.
- Equal weights are not balanced.
- Selected operating point: α:β:γ = 1:4:4.

Changing weights changes both performance and dispatch geography.

Weight choices reshape dispatch patterns

Policy	α:β:γ	Theil ↓	Cost ↓	Backup ↓
Cost-led	5:2:2	0.1137	647	115.76
Equal	1:1:1	0.1058	2,729	107.23
Balanced	**1:4:4**	**0.0931**	**9,854**	**98.06**
Equity-led	1:10:5	0.0760	13,820	105.30



Tri-CORE sits between localized, broad, and concentrated extremes-producing balanced redistribution.

03 WHAT THIS WORK CONTRIBUTES

01 Reveals the failure mode

Single-objective dispatch causes up to +97.7% inequality or +19.6% backup deficit.

02 Makes priorities explicit

One normalized objective turns α, β, and γ into interpretable policy priorities.

03 Finds a balanced policy

The 1:4:4 setting improves equity and resilience with only +0.3% distance.

TAKEAWAY

The dispatch trilemma is unavoidable.
But Tri-CORE makes it measurable, visible, and manageable.

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